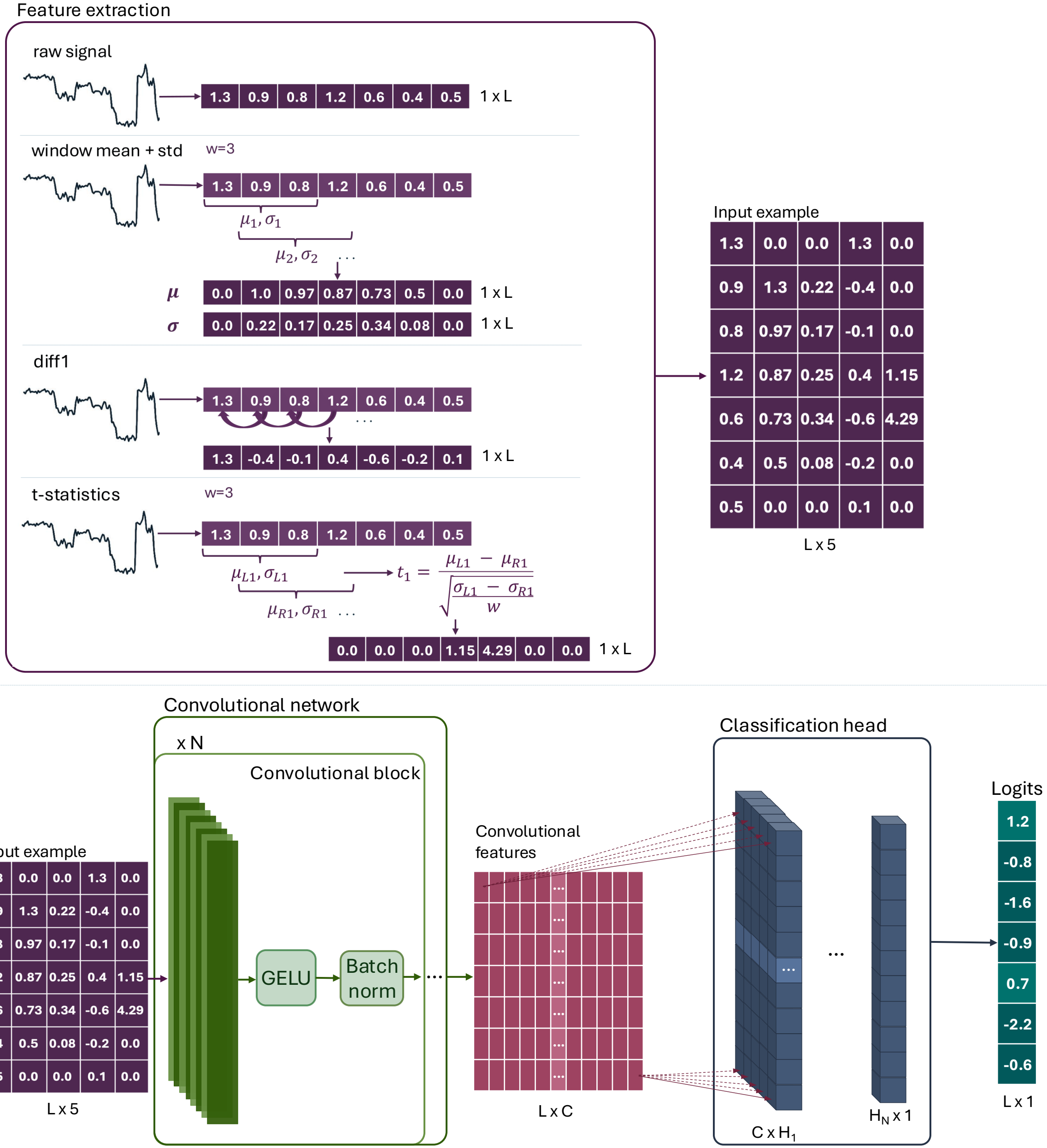
Feature extraction
raw signal
1.3 0.9 0.8 1.2 0.6 0.4 0.5
1 x L
window mean + std
w=3
1.3 0.9 0.8 1.2 0.6 0.4 0.5
μ_1, σ_1
μ_2, σ_2 ...
μ 0.0 1.0 0.97 0.87 0.73 0.5 0.0 1 x L
σ 0.0 0.22 0.17 0.25 0.34 0.08 0.0 1 x L
diff1
1.3 0.9 0.8 1.2 0.6 0.4 0.5
...
1.3 -0.4 -0.1 0.4 -0.6 -0.2 0.1 1 x L
t-statistics
w=3
1.3 0.9 0.8 1.2 0.6 0.4 0.5
μ_{L1}, σ_{L1}
μ_{R1}, σ_{R1} ...
$t_1 = \frac{\mu_{L1} - \mu_{R1}}{\sqrt{\frac{\sigma_{L1} - \sigma_{R1}}{w}}}$
0.0 0.0 0.0 1.15 4.29 0.0 0.0 1 x L
Input example
1.3 0.0 0.0 1.3 0.0
0.9 1.3 0.22 -0.4 0.0
0.8 0.97 0.17 -0.1 0.0
1.2 0.87 0.25 0.4 1.15
0.6 0.73 0.34 -0.6 4.29
0.4 0.5 0.08 -0.2 0.0
0.5 0.0 0.0 0.1 0.0
L x 5
Convolutional network
x N
Convolutional block
GELU
Batch norm
Convolutional features
L x C
Classification head
C x H_1
H_N x 1
Logits
1.2
-0.8
-1.6
-0.9
0.7
-2.2
-0.6
L x 1